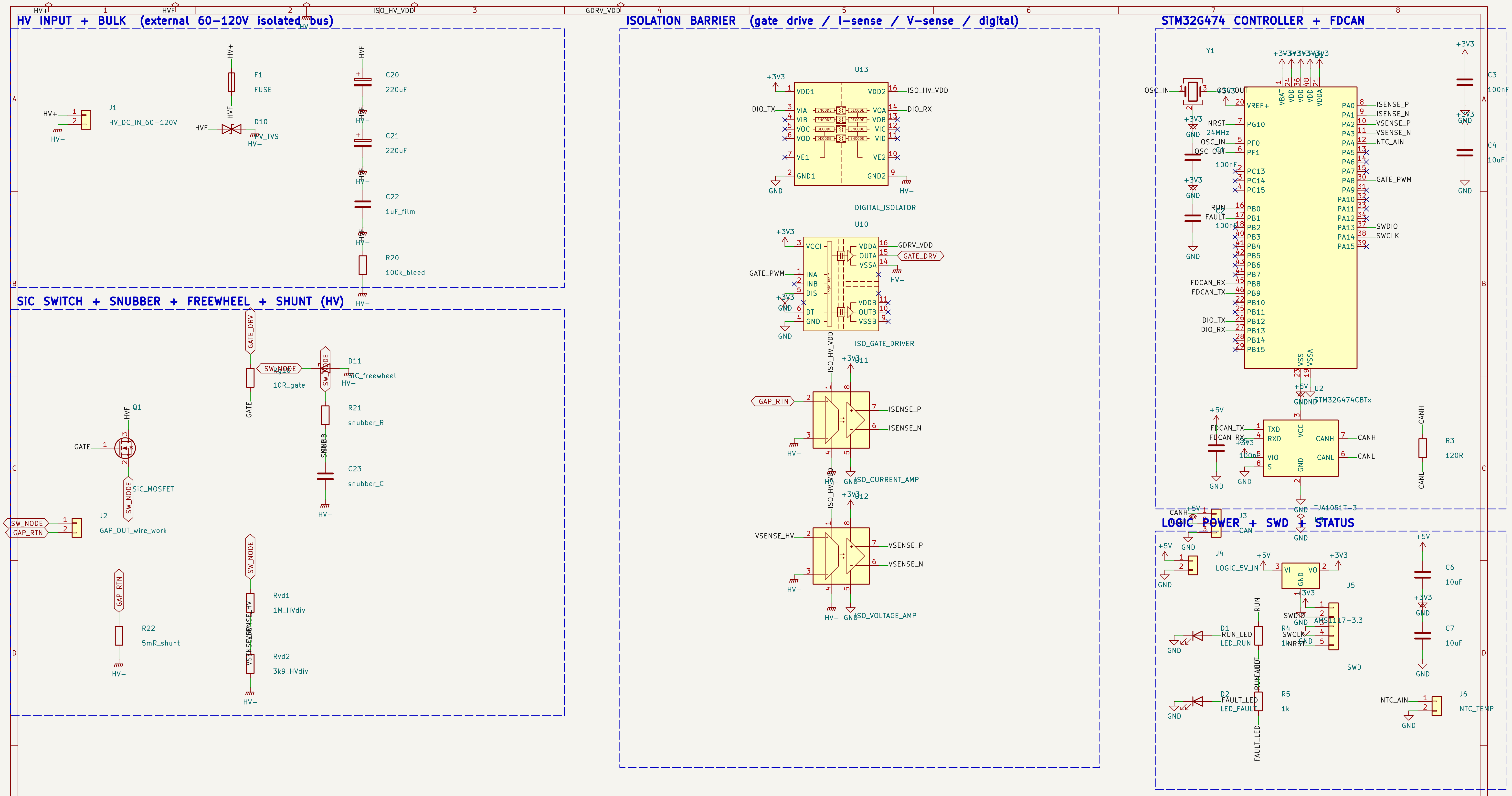
HV INPUT + BULK (external 60–120V isolated bus)

SIC SWITCH + SNUBBER + FREEWHEEL + SHUNT (HV)

ISOLATION BARRIER (gate drive / I–sense / V–sense / digital)

STM32G474 CONTROLLER + FDCAN

LOGIC POWER + SWD + STATUS

HV+
HVF
HV–
ISO_HV_VDD
GDRV_VDD
J1
HV_DC_IN_60–120V
F1
FUSE
D10
HV_TVS
C20
220uF
C21
220uF
C22
1uF_film
R20
100k_bleed
GATE_DRV
SW_NODE
Rg10
10R_gate
GATE
D11
SiC_freewheel
R21
snubber_R
SNUB
C23
snubber_C
Q1
SiC_MOSFET
J2
GAP_OUT_wire_work
GAP_RTN
R22
5mR_shunt
Rvd1
1M_HVdiv
VSENSE_HV
Rvd2
3k9_HVdiv
U13
+3V3
VDD1
VDD2
DIO_TX
DIO_RX
VIA
VIB
VOC
VOD
VOA
VOB
VIC
VID
VE1
VE2
GND1
GND2
ENCODE
DECODE
GND
DIGITAL_ISOLATOR
U10
VCCI
VDDA
OUTA
VSSA
GATE_PWM
INA
INB
DIS
DT
VDDB
OUTB
VSSB
Logic Input
ISO_GATE_DRIVER
U11
ISENSE_P
ISENSE_N
ISO_CURRENT_AMP
U12
VSENSE_P
VSENSE_N
ISO_VOLTAGE_AMP
Y1
24MHz
OSC_IN
OSC_OUT
C1
100nF
C2
100nF
NRST
VREF+
PG10
PF0
PF1
PC13
PC14
PC15
RUN
FAULT
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
FDCAN_RX
FDCAN_TX
VBAT
VDD
VDDA
VSS
VSSA
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
NTC_AIN
SWDIO
SWCLK
U1
U2
STM32G474CBTx
C3
100nF
C4
10uF
+5V
C5
100nF
TXD
RXD
VIO
S
VCC
CANH
CANL
TJA1051T–3
U3
R3
120R
J3
CAN
J4
LOGIC_5V_IN
VI
VO
AMS1117–3.3
J5
SWD
NRST
D1
RUN_LED
LED_RUN
R4
1k
D2
FAULT_LED
LED_FAULT
R5
1k
C6
10uF
C7
10uF
J6
NTC_TEMP
1
2
3
4
5
6
7
8
A
B
C
D
E
F

Microspark
Sheet: /
File: MS–EDM–PSU.kicad_sch

Title: MS–EDM–PSU EDM Pulse Supply [HV – REVIEW ONLY, NOT FAB–

Size: A3 Date: 2026–06–16 **Rev: A**
KiCad E.D.A. 10.0.3 Id: 1/1